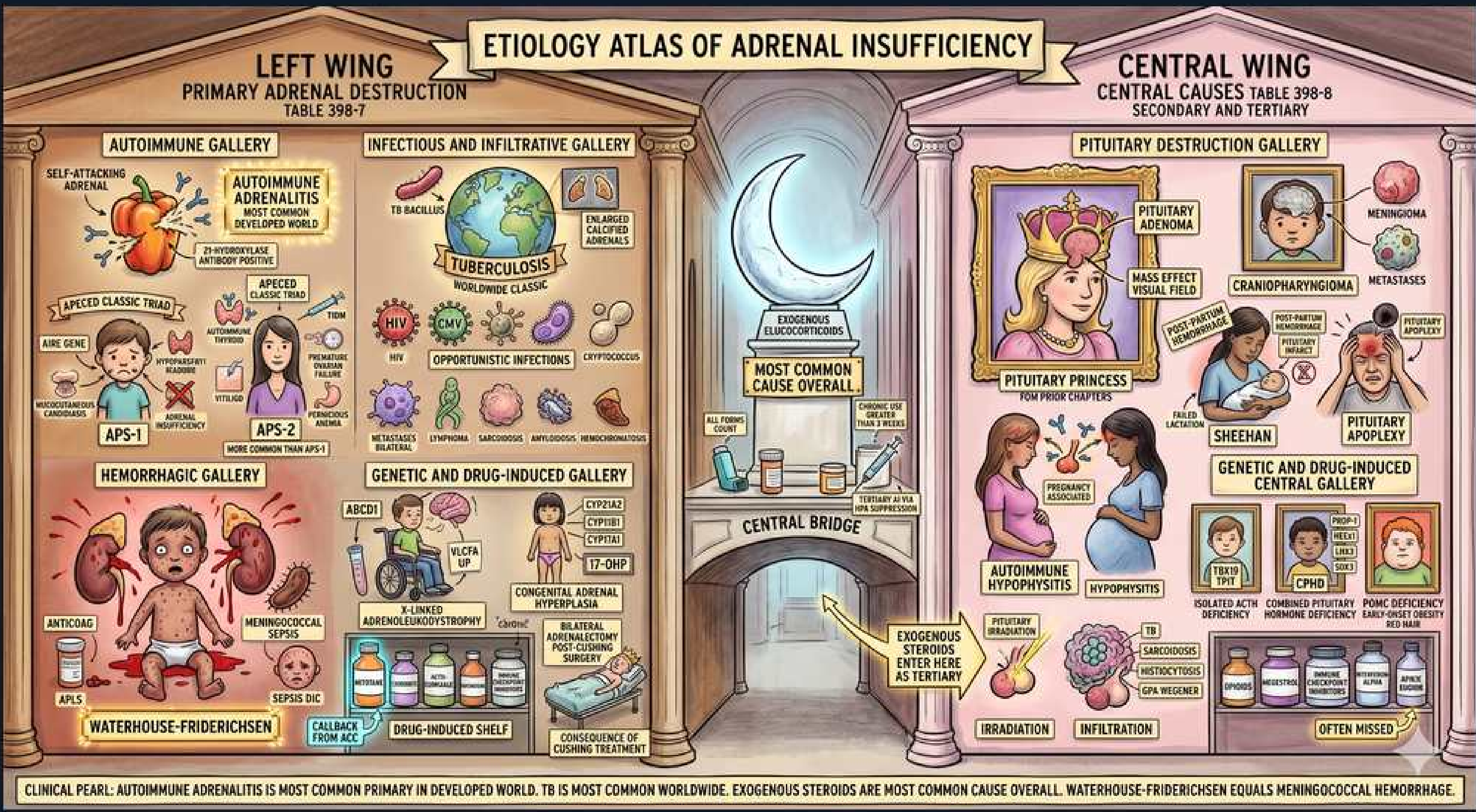
Adrenal Insufficiency/CAH — Etiology Atlas of Adrenal Insufficiency



1. Left wing — primary causes

WHAT YOU SEE

Banner LEFT WING: primary adrenal destruction

WHAT IT ENCODES

The left wing collects causes of PRIMARY AI, where the adrenal cortex itself is destroyed or fails. Because cortisol AND aldosterone AND androgens all fall, the labs show low cortisol, low aldosterone with high renin, and high ACTH (lost feedback). The high ACTH/MSH causes hyperpigmentation, and aldosterone loss causes hyponatremia + hyperkalemia. This is the only category with mineralocorticoid loss.

BOARD TRAP

Primary AI is the ONLY type with high ACTH, low aldosterone, hyperkalemia, and hyperpigmentation together.

---

2. Autoimmune adrenalitis

WHAT YOU SEE

Self-attacking adrenal, AUTOIMMUNE ADRENALITIS, most common in developed world

WHAT IT ENCODES

Autoimmune (Addison) adrenalitis is the #1 cause of primary AI in the developed world, driven by 21-hydroxylase autoantibodies that destroy the cortex. It can be isolated or part of a polyglandular syndrome. The zona glomerulosa is often hit early, so isolated hyperreninemia/aldosterone loss can precede overt cortisol deficiency.

BOARD TRAP

Autoimmune is #1 in DEVELOPED countries, but TB remains #1 WORLDWIDE — the board loves this geography swap.

---

3. APS-1 / APS-2

WHAT YOU SEE

APS-1 and APS-2 polyglandular syndrome panels

WHAT IT ENCODES

Autoimmune AI clusters in autoimmune polyglandular syndromes. APS-1 (AIRE gene, childhood): the triad of chronic mucocutaneous candidiasis + hypoparathyroidism + Addison disease (CMC + HypoPTH + Adrenal). APS-2 (adults, more common): Addison + autoimmune thyroid disease and/or type 1 diabetes (Schmidt syndrome). Screen patients with one autoimmune endocrinopathy for the others.

BOARD TRAP

APS-1 = candidiasis + hypoparathyroidism + Addison; APS-2 = Addison + thyroid + T1DM (no candidiasis/hypoparathyroidism).

---

4. Tuberculosis

WHAT YOU SEE

Globe labeled TUBERCULOSIS, worldwide classic

WHAT IT ENCODES

TB is the most common cause of primary AI WORLDWIDE, producing bilateral infiltrative/granulomatous destruction (and sometimes adrenal calcification on CT). It destroys both cortex and medulla over time. In endemic regions or immigrants, work up new primary AI with a chest X-ray and adrenal imaging.

BOARD TRAP

TB (not autoimmune) is #1 worldwide; bilateral adrenal calcification on imaging points to TB or prior hemorrhage.

---

5. Infiltrative / infectious gallery

WHAT YOU SEE

Infectious and infiltrative gallery (fungi, HIV-related, etc.)

WHAT IT ENCODES

Other primary causes include opportunistic/disseminated infections (HIV-associated CMV adrenalitis, histoplasmosis, other fungi), metastatic disease (lung, breast — adrenals are a common metastatic site), and infiltrative diseases (amyloidosis, hemochromatosis, sarcoidosis). These typically need bilateral involvement to cause AI because of large cortical reserve.

BOARD TRAP

Adrenal metastases are common but rarely cause AI unless BOTH glands are largely replaced (>90% cortex lost).

---

6. Waterhouse-Friderichsen

WHAT YOU SEE

Hemorrhagic gallery, baby with meningococcal sepsis, bilateral adrenal hemorrhage

WHAT IT ENCODES

Waterhouse-Friderichsen syndrome is acute bilateral adrenal hemorrhage classically from Neisseria meningitidis sepsis (also Pseudomonas, DIC). It presents as fulminant shock, purpura, and acute primary AI (adrenal crisis). Anticoagulation, antiphospholipid syndrome, and severe stress/surgery are other causes of bilateral adrenal hemorrhage.

BOARD TRAP

Waterhouse-Friderichsen = meningococcemia + bilateral adrenal hemorrhage → acute adrenal crisis; give stress-dose hydrocortisone immediately, don't wait for confirmatory labs.

---

7. Genetic / drug-induced (primary)

WHAT YOU SEE

Genetic and drug-induced shelf, CAH, adrenoleukodystrophy, ketoconazole, etc.

WHAT IT ENCODES

Genetic primary causes: congenital adrenal hyperplasia (21-hydroxylase deficiency, ~90% of CAH — low cortisol/aldosterone with high androgens and salt-wasting) and X-linked adrenoleukodystrophy (ABCD1, very-long-chain fatty acids; AI + neurologic decline in boys/men). Drugs that block adrenal steroidogenesis: ketoconazole, etomidate (single dose can drop cortisol), metyrapone, mitotane.

BOARD TRAP

In a young man with AI plus progressive neurologic/behavioral decline, think X-linked adrenoleukodystrophy (check very-long-chain fatty acids).

---

8. Central wing — secondary/tertiary

WHAT YOU SEE

Banner CENTRAL WING: central causes, secondary and tertiary

WHAT IT ENCODES

The central wing holds CENTRAL AI: secondary (pituitary ACTH deficiency) and tertiary (hypothalamic CRH deficiency, including steroid suppression). ACTH is low/inappropriately normal, so cortisol and androgens fall but ALDOSTERONE IS PRESERVED (RAAS-driven). Result: no hyperkalemia, no salt craving, no hyperpigmentation (low ACTH/MSH), pale skin.

BOARD TRAP

Central AI keeps aldosterone → normal potassium and pale skin; only cortisol-deficiency hyponatremia (via ADH) is seen.

---

9. Pituitary tumor / mass effect

WHAT YOU SEE

Pituitary princess with mass-effect/visual-field damage, craniopharyngioma

WHAT IT ENCODES

Pituitary macroadenomas, craniopharyngiomas, and metastases destroy or compress corticotrophs → secondary AI, often with mass effect (bitemporal hemianopia from optic chiasm compression) and other hormone deficits (panhypopituitarism). ACTH is typically among the LAST axes lost (order often: GH → gonadotropins → TSH → ACTH).

BOARD TRAP

In hypopituitarism, give glucocorticoid BEFORE levothyroxine — starting thyroid hormone first can precipitate adrenal crisis.

---

10. Sheehan syndrome

WHAT YOU SEE

Sheehan panel (postpartum pituitary infarct)

WHAT IT ENCODES

Sheehan syndrome is postpartum pituitary necrosis from peripartum hemorrhage/hypotension acting on the gestationally enlarged gland → secondary AI plus other deficits. Classic clues: failure to lactate (prolactin loss) and failure to resume menses (gonadotropin loss), later fatigue/hypothyroid features. Onset can be acute or insidious months to years later.

BOARD TRAP

Postpartum woman who can't lactate and has amenorrhea → Sheehan; remember her acute AI may still pass a cosyntropin test if the adrenal hasn't yet atrophied.

---

11. Pituitary apoplexy

WHAT YOU SEE

Pituitary apoplexy panel (acute hemorrhage)

WHAT IT ENCODES

Pituitary apoplexy is sudden hemorrhage or infarction of the pituitary (often into a pre-existing adenoma) → thunderclap headache, ophthalmoplegia, visual loss, and ACUTE secondary AI. It is an endocrine/neurosurgical emergency requiring immediate stress-dose glucocorticoids and urgent imaging.

BOARD TRAP

Sudden severe headache + visual change + hypotension in a known/occult pituitary adenoma = apoplexy → give hydrocortisone STAT before imaging confirms.

---

12. Exogenous steroids (tertiary)

WHAT YOU SEE

Central bridge: EXOGENOUS GLUCOCORTICOIDS, most common cause overall, enter as tertiary

WHAT IT ENCODES

Chronic exogenous glucocorticoids suppress hypothalamic CRH and pituitary ACTH (tertiary AI) and cause the adrenal to atrophy — this is the MOST COMMON cause of AI overall. Risk rises with dose, duration (>3 weeks of supraphysiologic dosing), and potency; abrupt withdrawal or an intercurrent stress unmasks crisis. Always taper and stress-dose.

BOARD TRAP

Any prednisone equivalent >20 mg/day for >3 weeks (or Cushingoid appearance) suppresses the axis — never stop abruptly; taper and stress-dose for illness/surgery.

---

13. Genetic/infiltrative central

WHAT YOU SEE

Genetic and drug-induced central gallery (isolated ACTH deficiency, CPHD, sarcoid, irradiation, infiltration)

WHAT IT ENCODES

Central AI also arises from congenital combined pituitary hormone deficiency (CPHD), isolated ACTH deficiency, cranial irradiation, traumatic brain injury, and infiltrative disease — lymphocytic hypophysitis, sarcoidosis, hemochromatosis, and Langerhans cell histiocytosis. Newer cause: immune checkpoint inhibitors causing hypophysitis (secondary) or adrenalitis (primary).

BOARD TRAP

Checkpoint inhibitors (ipilimumab/nivolumab) are an increasingly tested cause — hypophysitis → secondary AI; direct adrenalitis → primary AI.

---

14. Clinical pearl banner

WHAT YOU SEE

Bottom pearl: autoimmune #1 primary developed, TB worldwide, exogenous steroids most common overall

WHAT IT ENCODES

Etiology one-liner to memorize: autoimmune adrenalitis = #1 primary in developed countries; TB = #1 primary worldwide; Waterhouse-Friderichsen = meningococcal bilateral adrenal hemorrhage; exogenous glucocorticoids = #1 cause of AI overall (tertiary). Localize with ACTH (high = primary, low/normal = central) plus renin/aldosterone and 21-hydroxylase antibodies.

BOARD TRAP

Don't conflate 'most common primary' (autoimmune in developed world) with 'most common overall' (exogenous steroids).
